Data Sciences – CentraleSupelec
Large Scale and Distributed Optimization
Part IV: Proximity operator

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Motivation

Let $\mathcal{H}$ be a real Hilbert space. Let $f \in \Gamma_0(\mathcal{H})$ have a Lipschitz gradient with Lipschitz constant $\beta > 0$.
Find

$$\widehat{x} \in \operatorname*{Argmin}_{x \in \mathcal{H}} f(x).$$

- **Gradient descent algorithm**

  Set $\gamma \in \,]0, +\infty[$ and $x_0 \in \mathcal{H}$.
  For $n = 0, 1 \ldots$
  $\lfloor\ x_{n+1} = x_n - \gamma \nabla f(x_n)$.

The sequence $(x_n)_{n \in \mathbb{N}}$ generated by this *explicit* scheme converges to a minimizer of f provided that such a minimizer exists and $\gamma \in]0, 2/\beta[$.

Motivation

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Find

$$\widehat{x} \in \operatorname*{Argmin}_{x \in \mathcal{H}} f(x).$$

- **Alternative algorithm**

  Set $\gamma \in \left]0, +\infty\right[$ and $x_0 \in \mathcal{H}$.
  For $n = 0, 1 \ldots$
  $\lfloor\ x_{n+1} = x_n - \gamma \nabla f(x_{n+1})$.

Questions:

- How to determine x_{n+1} at each iteration n of this *implicit* scheme ?
- Which values of γ guarantee the convergence of $(x_n)_{n \in \mathbb{N}}$?
- What to do if f is nonsmooth ?

Proximity operator: definition

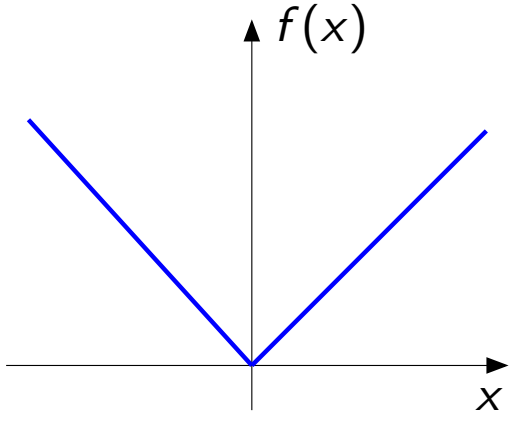
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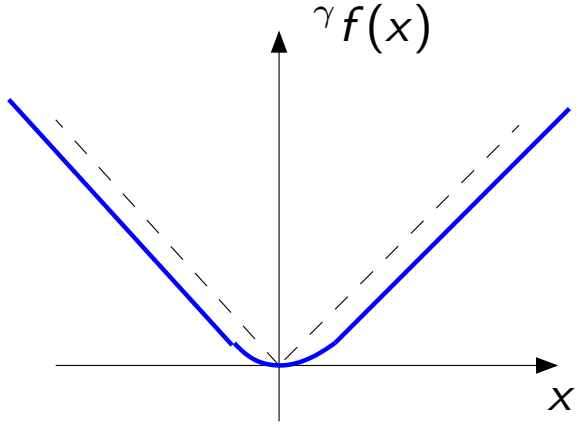
- The **Moreau envelope** of f of parameter $\gamma \in \,]0, +\infty[$ is

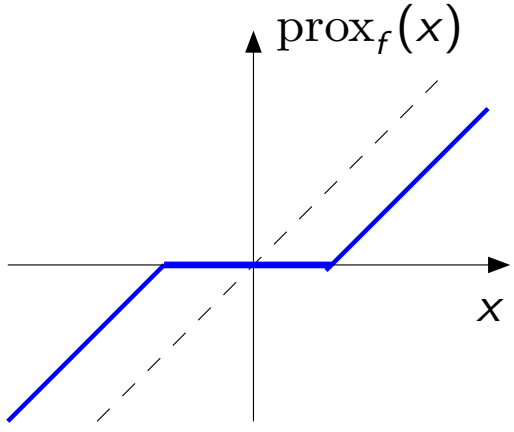
$$ {}^{\gamma}f : \mathcal{H} \to \mathbb{R} : x \mapsto \inf_{y \in \mathcal{H}} f(y) + \frac{1}{2\gamma}\|y - x\|^2. $$

- The **proximity operator** of f is

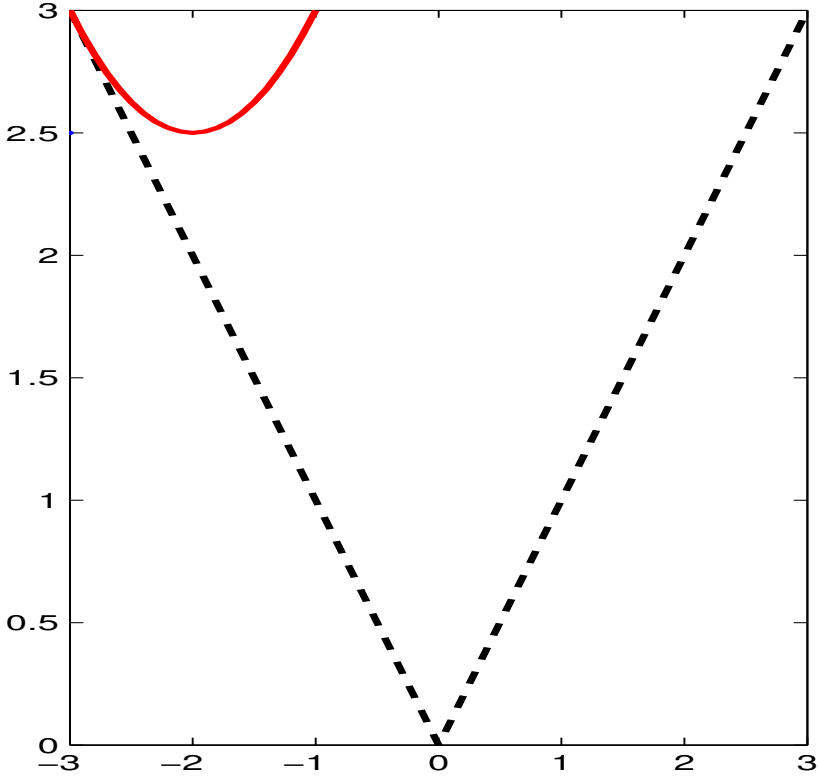
$$ \operatorname{prox}_f : \mathcal{H} \to \mathcal{H} : x \mapsto \operatorname*{argmin}_{y \in \mathcal{H}} f(y) + \frac{1}{2}\|y - x\|^2. $$





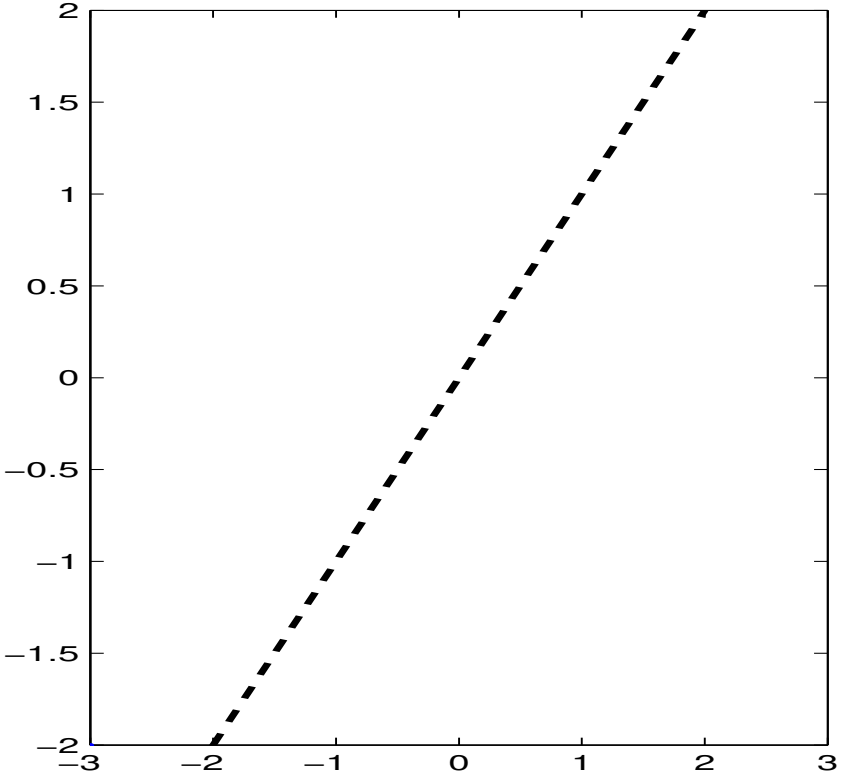


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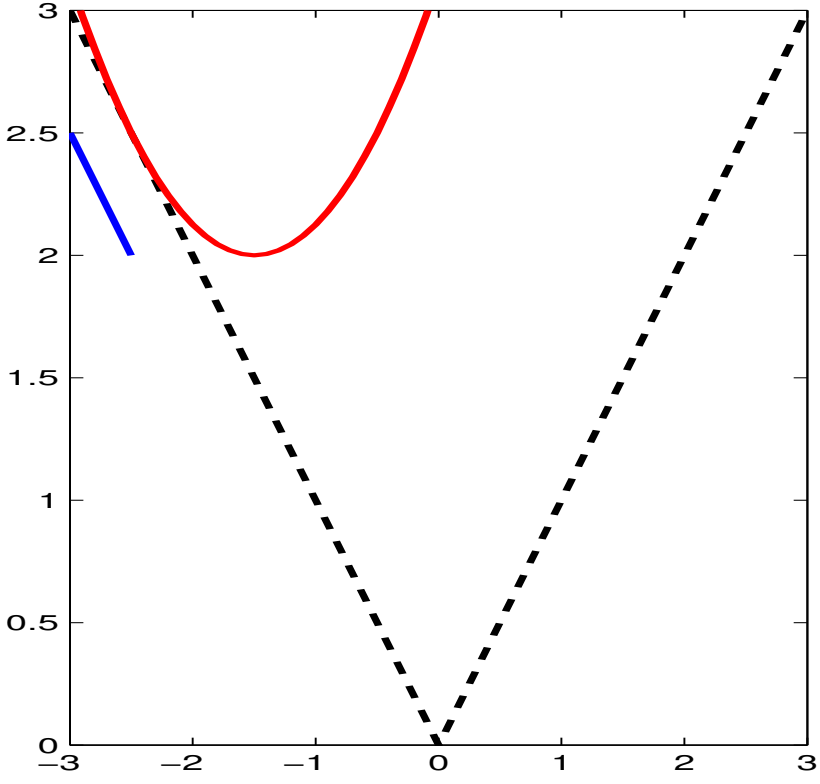
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Proximity operator

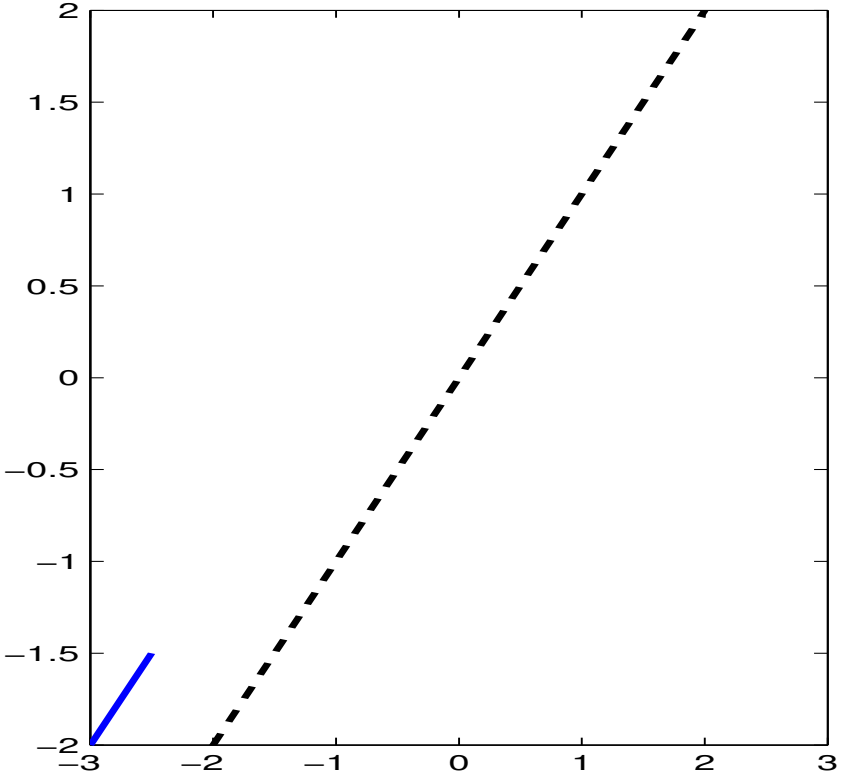
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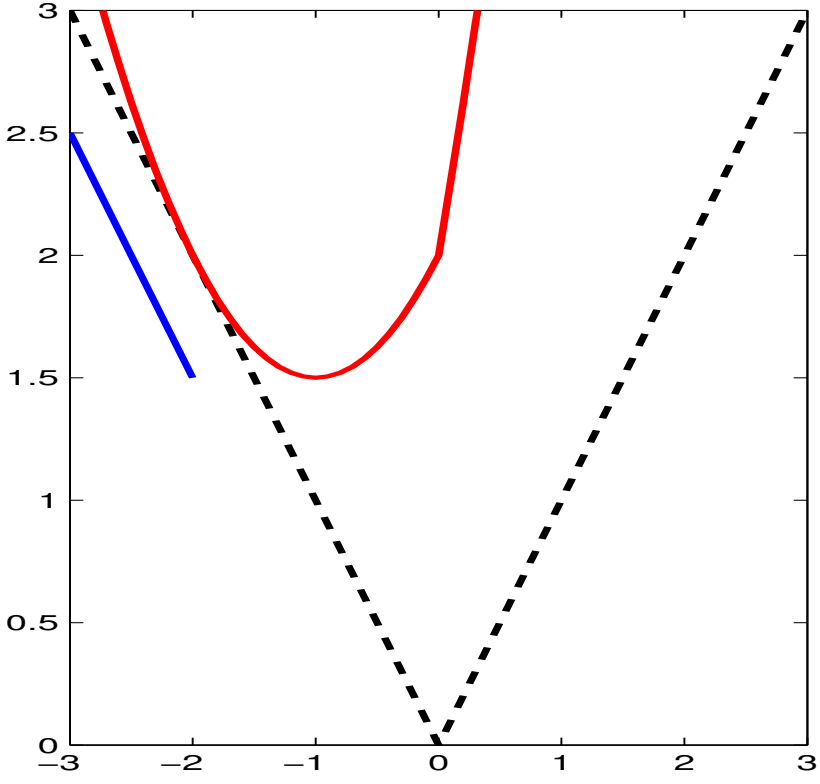
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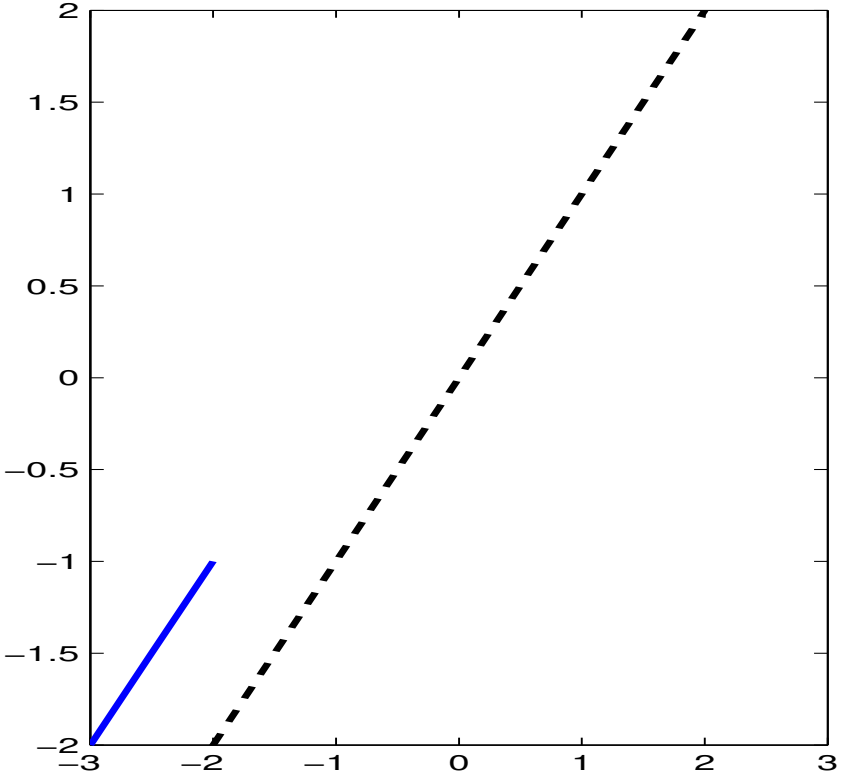
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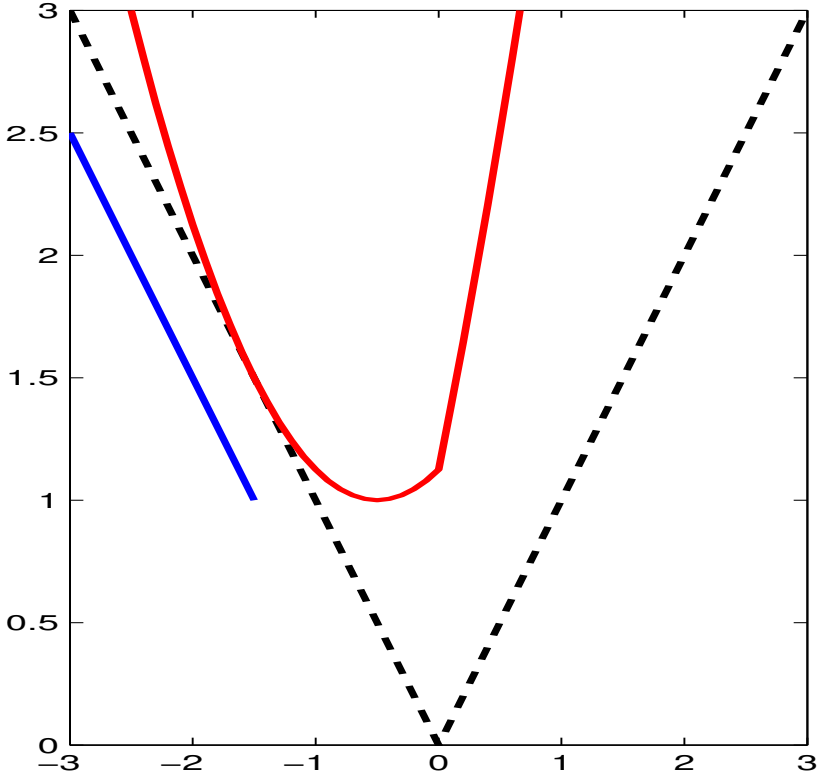
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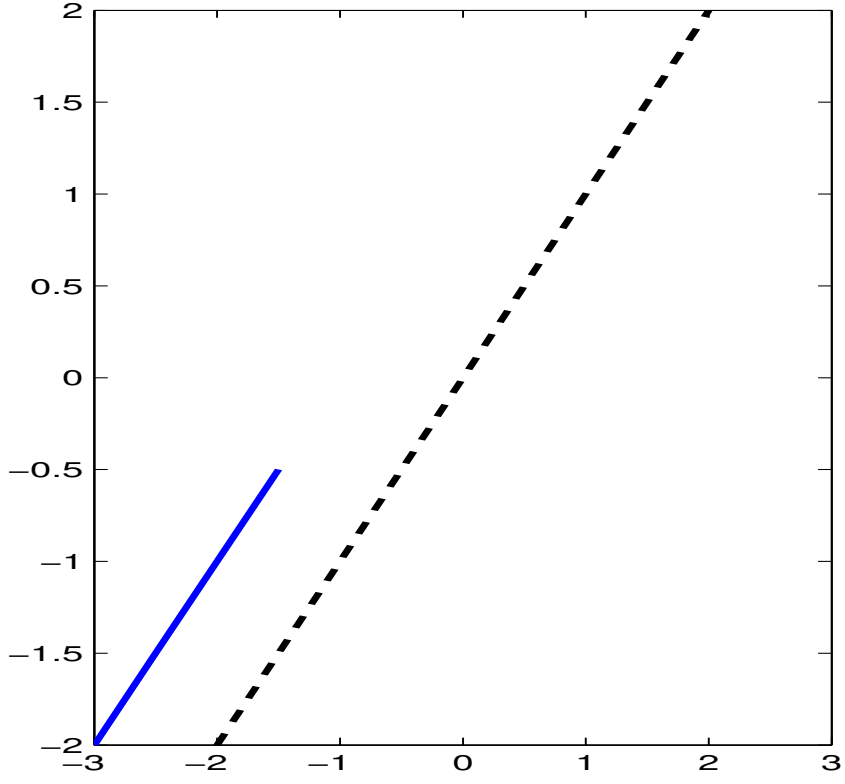
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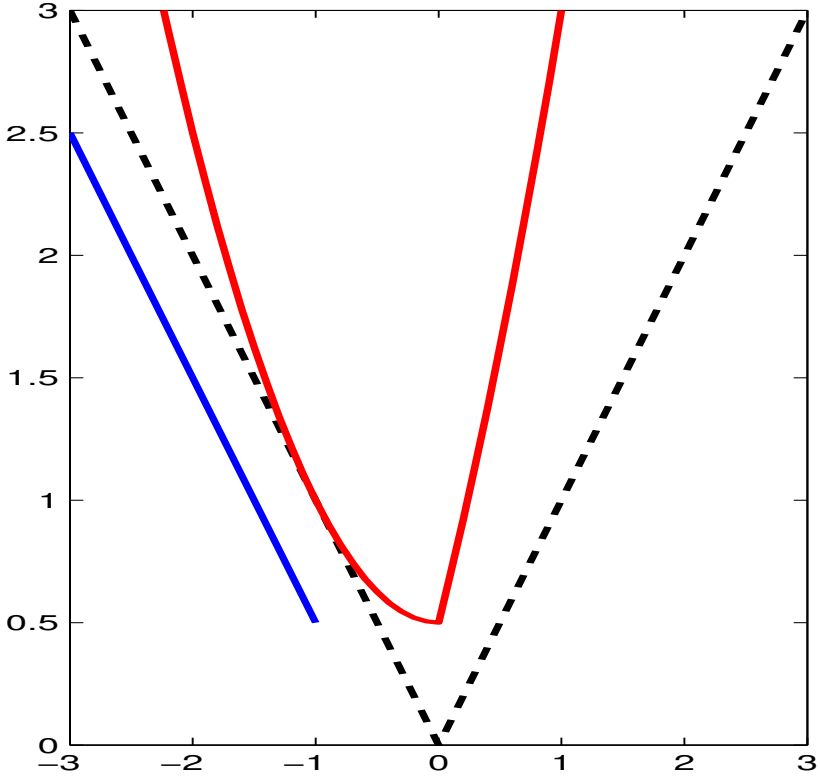
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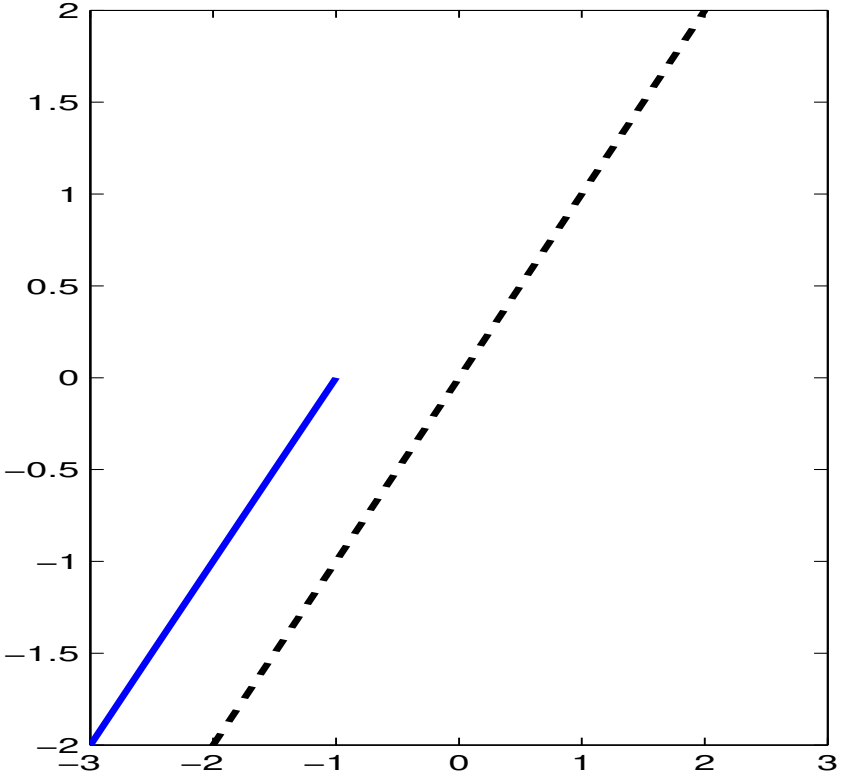
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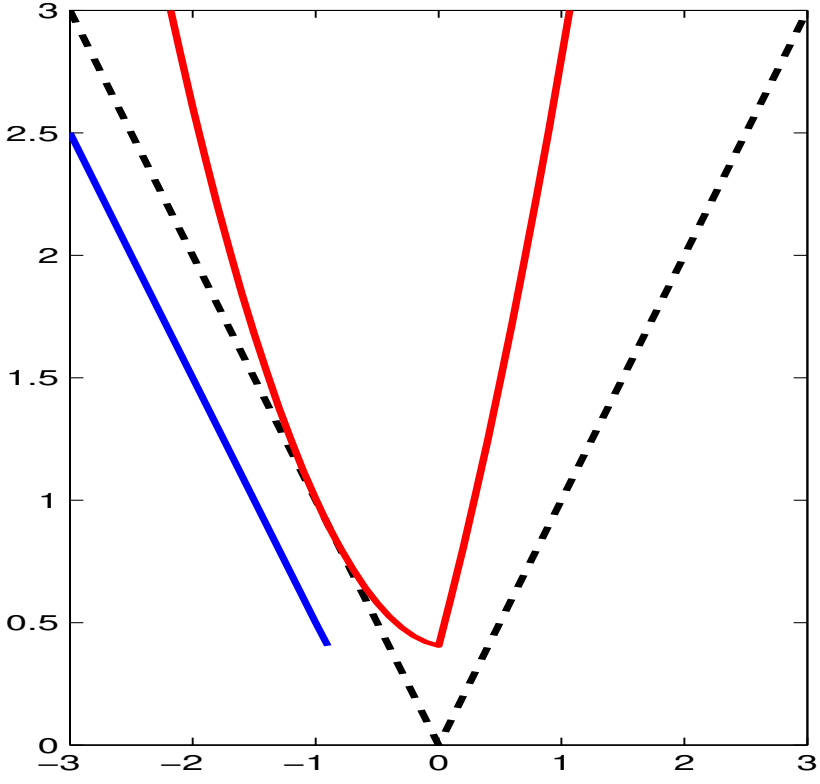
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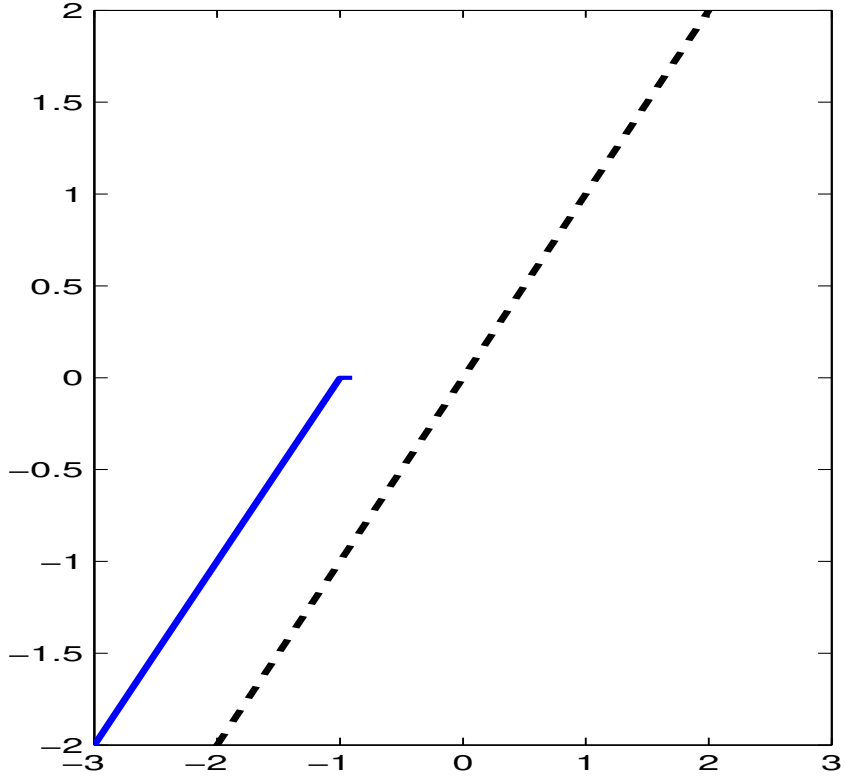
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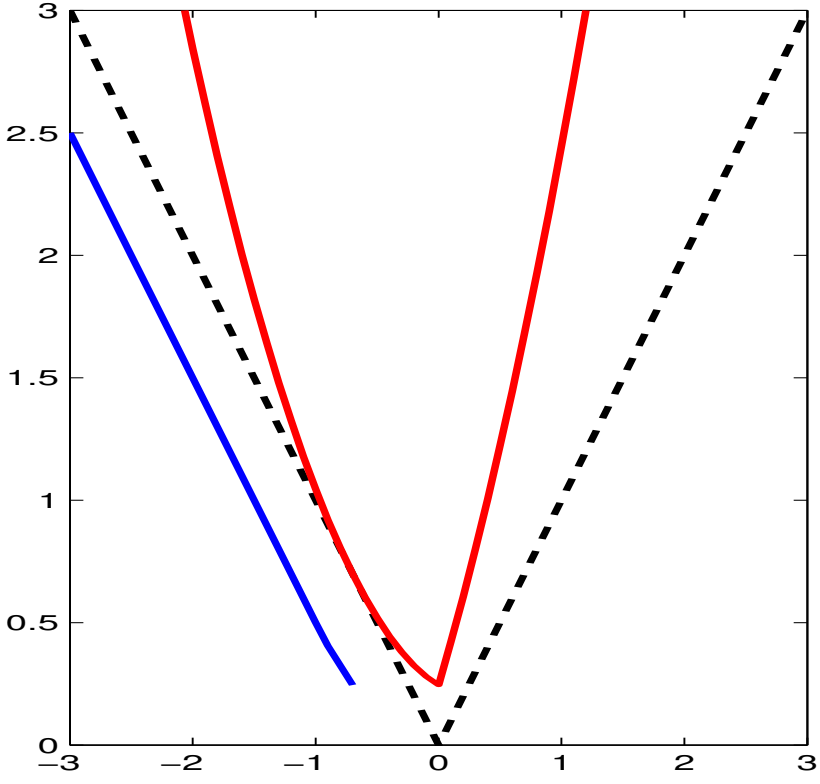
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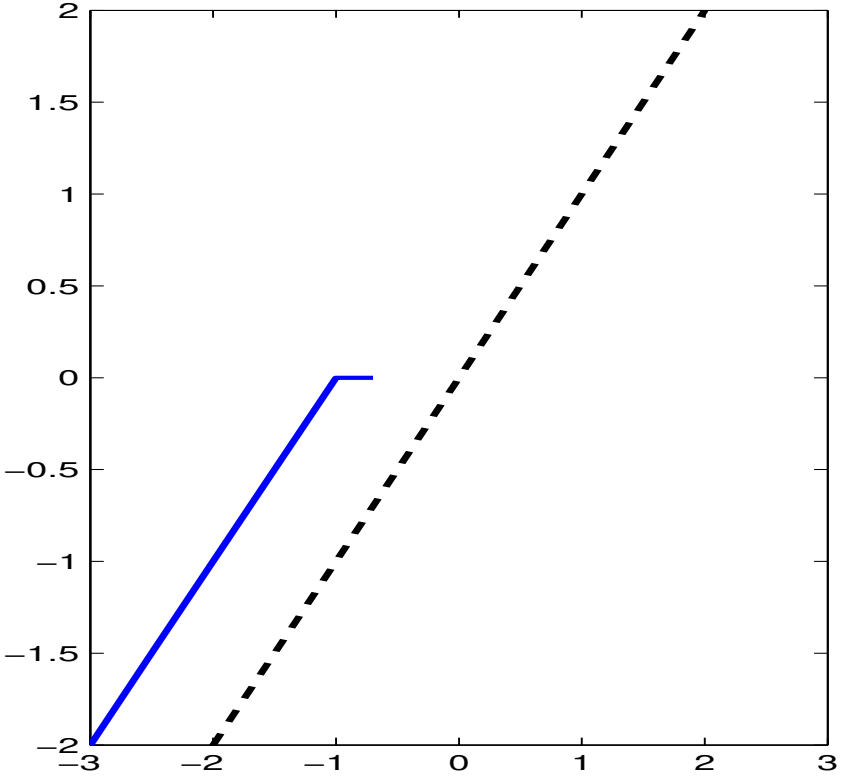
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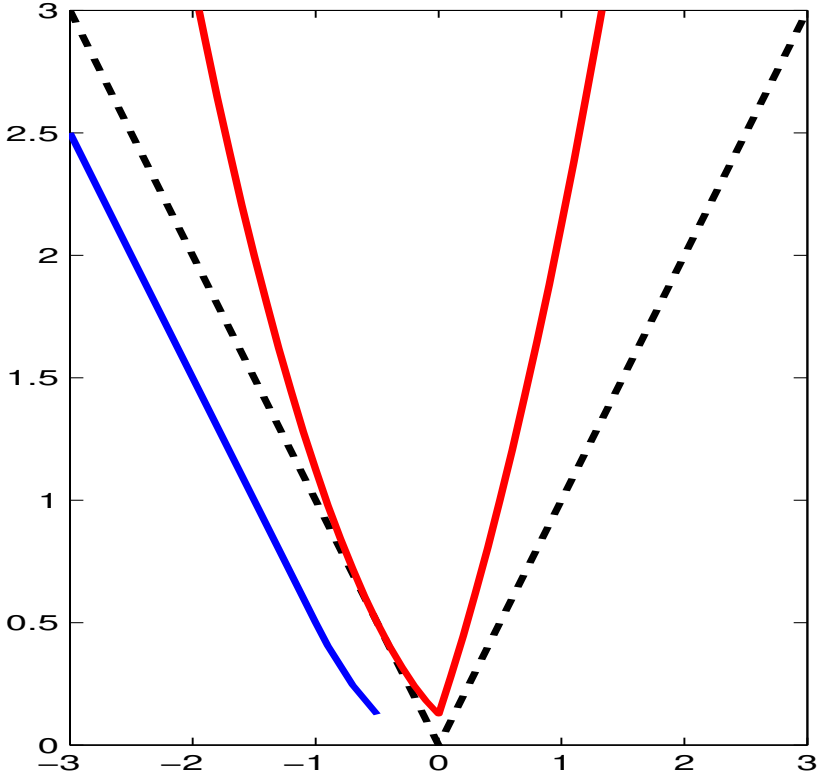
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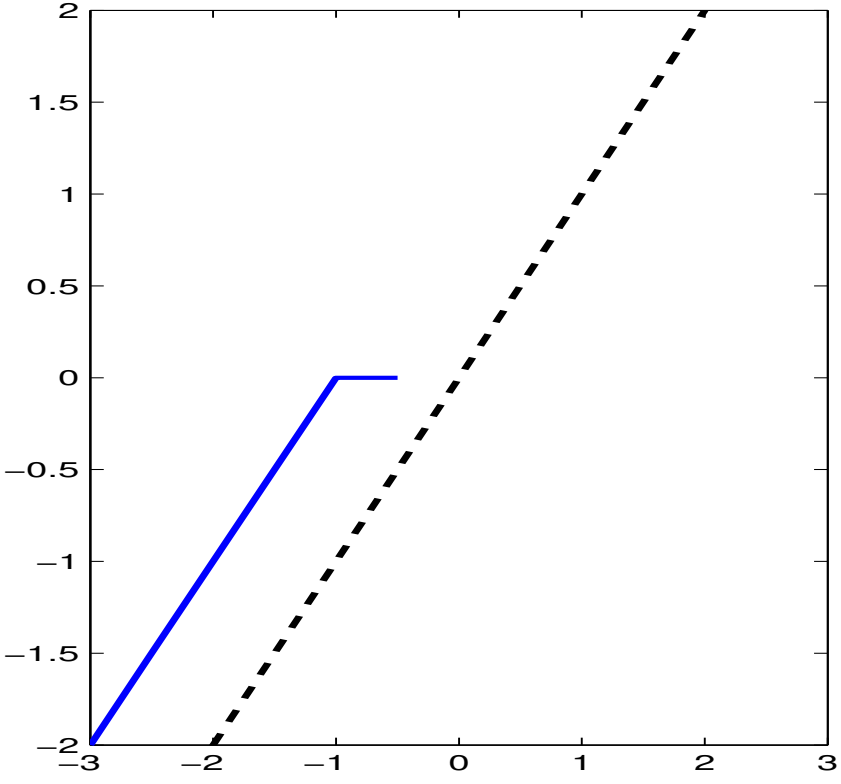
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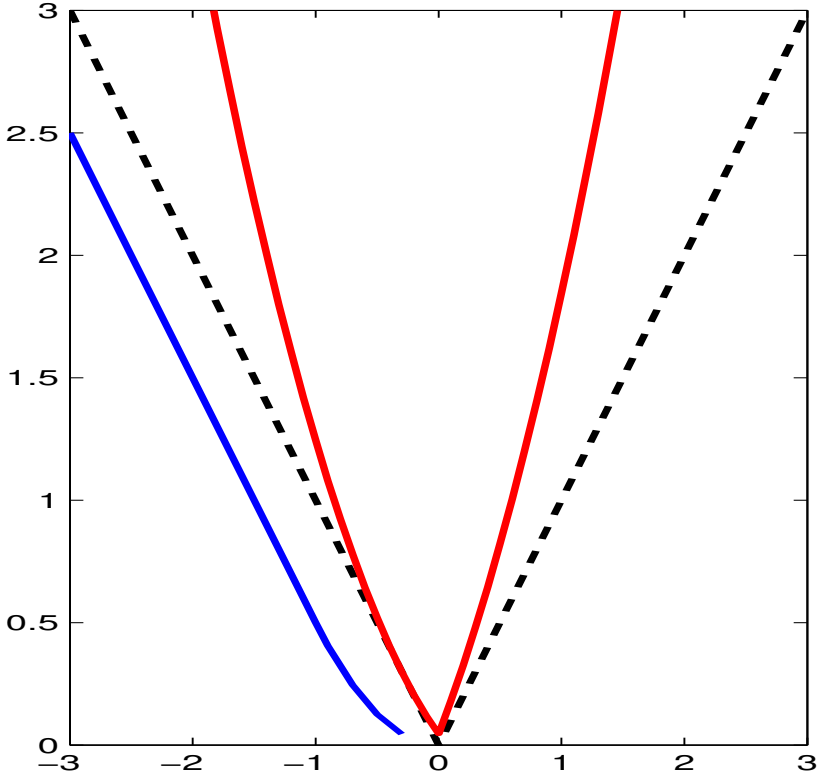
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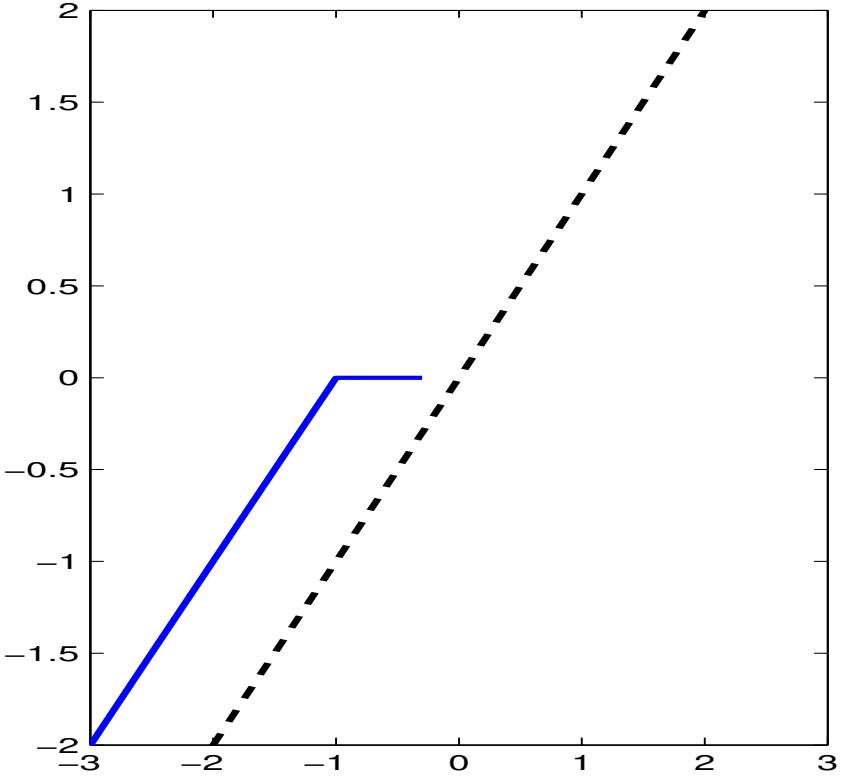
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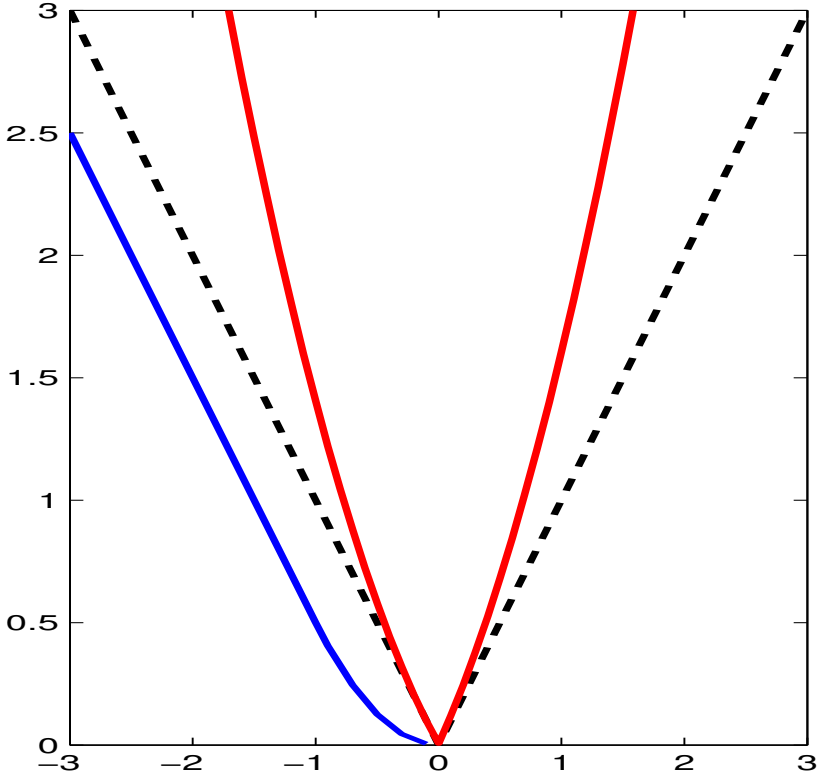
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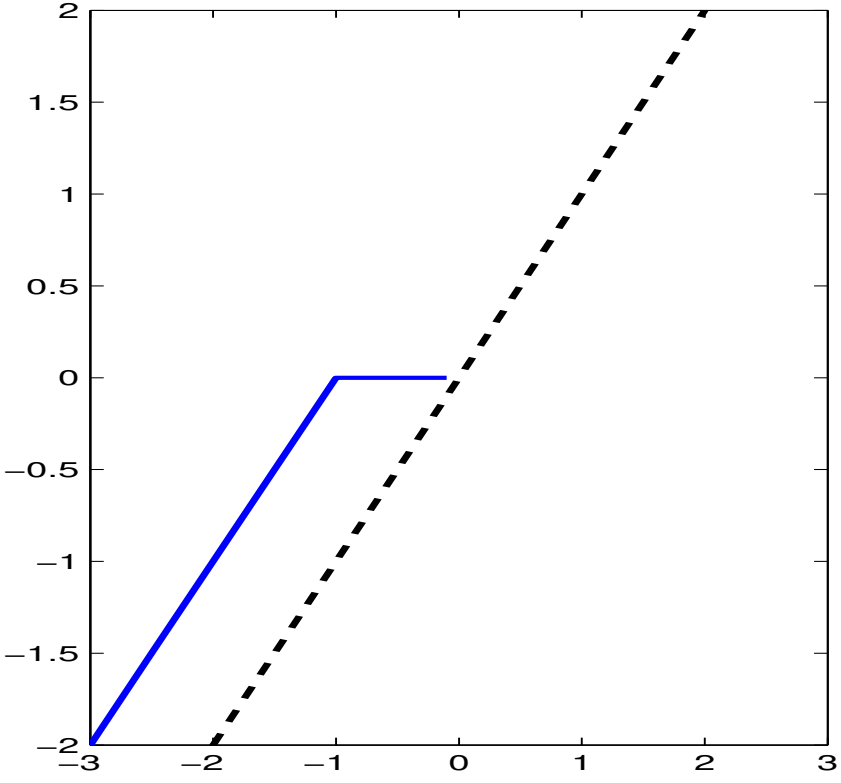
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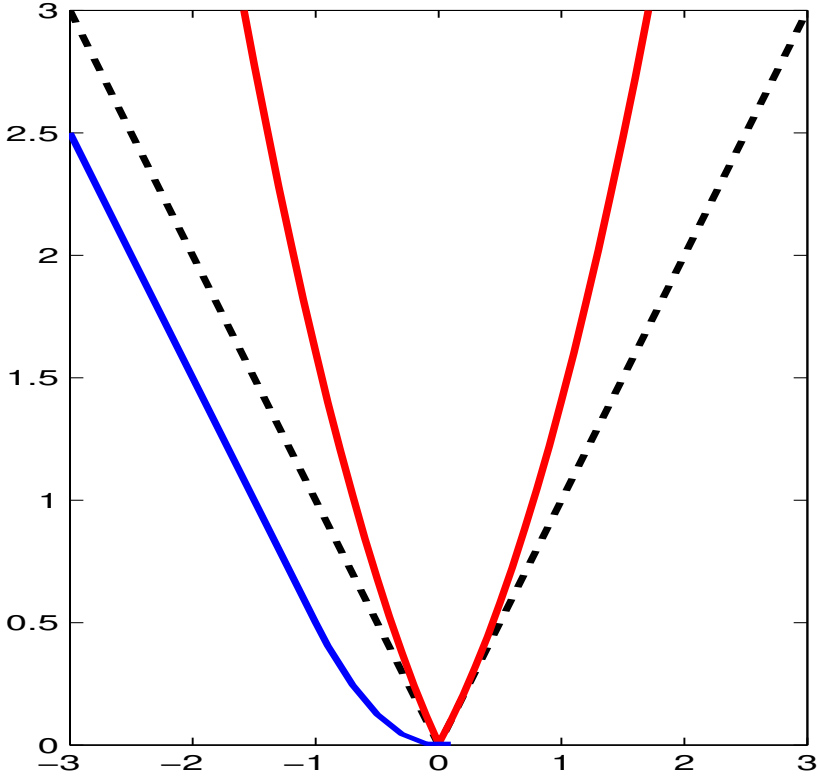
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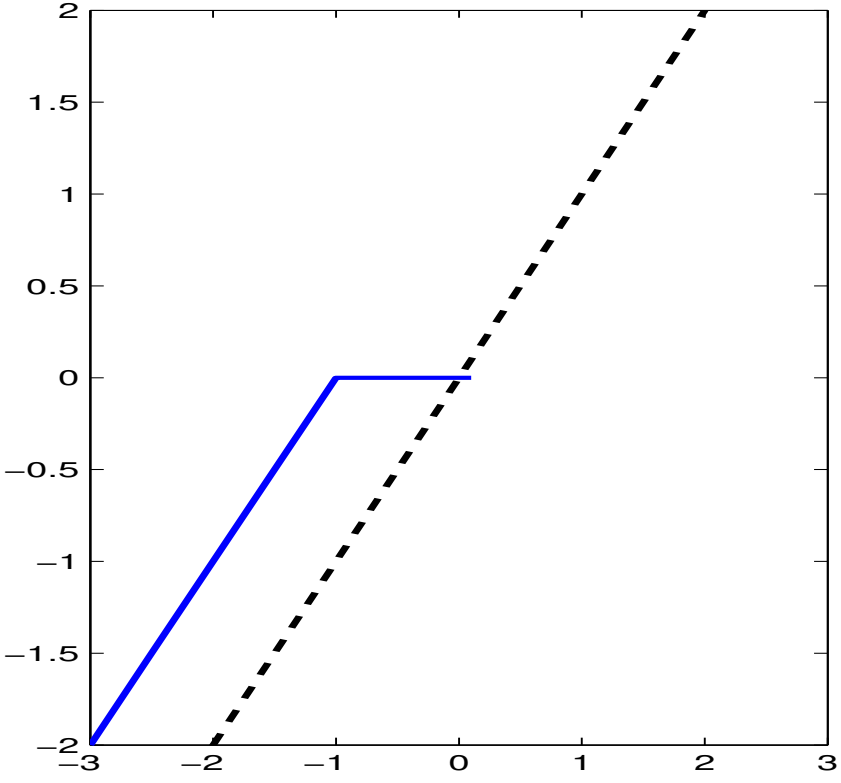
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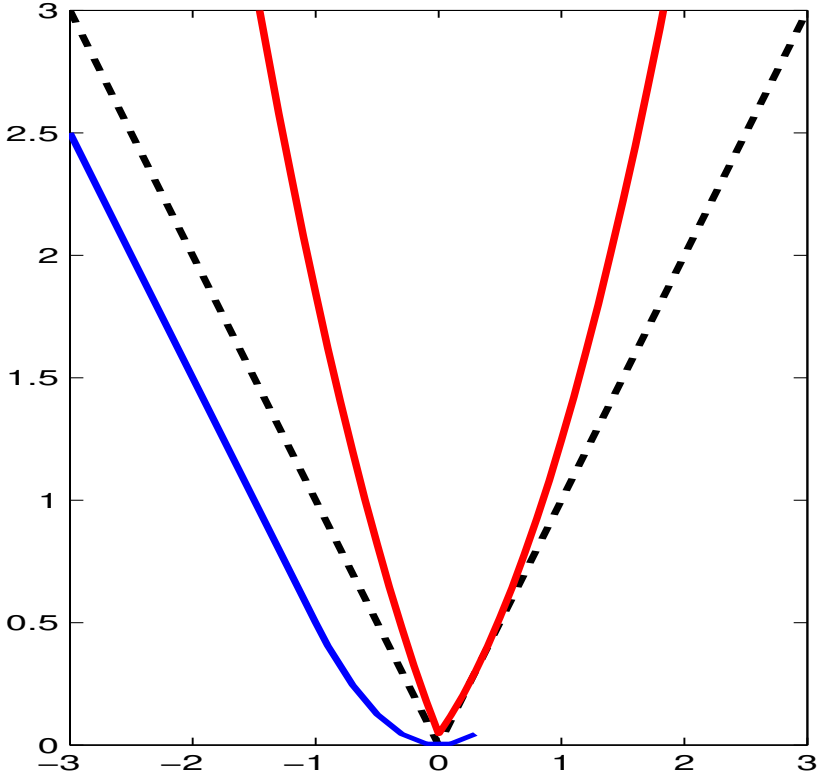
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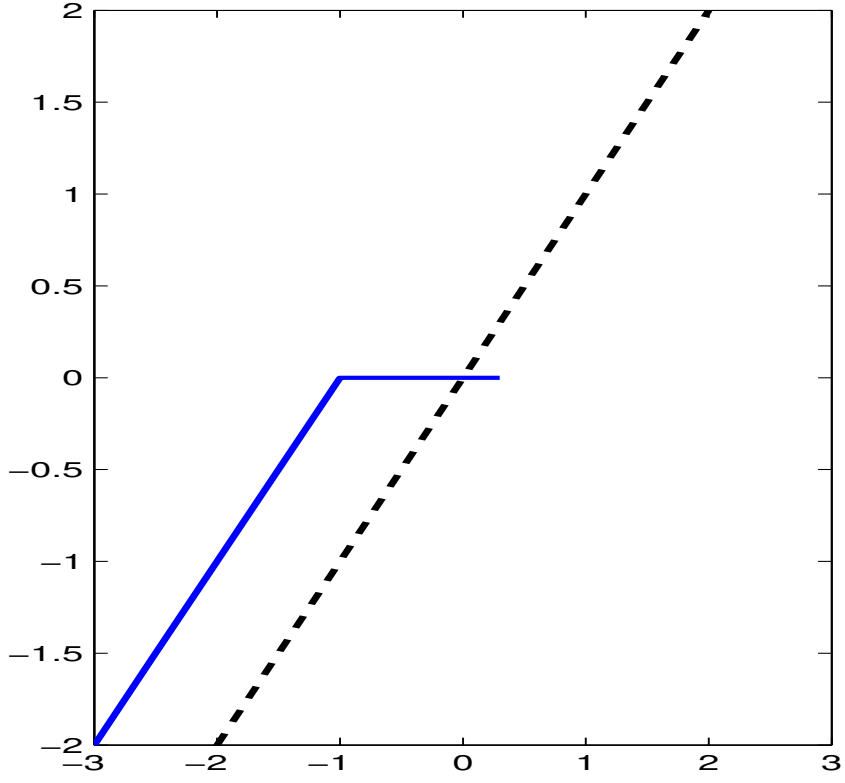
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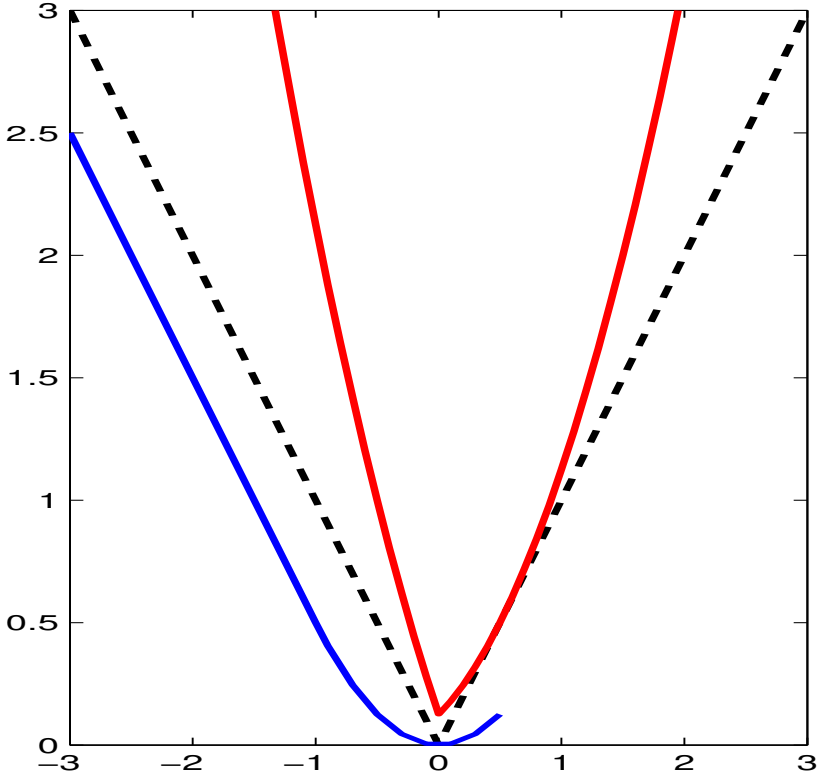
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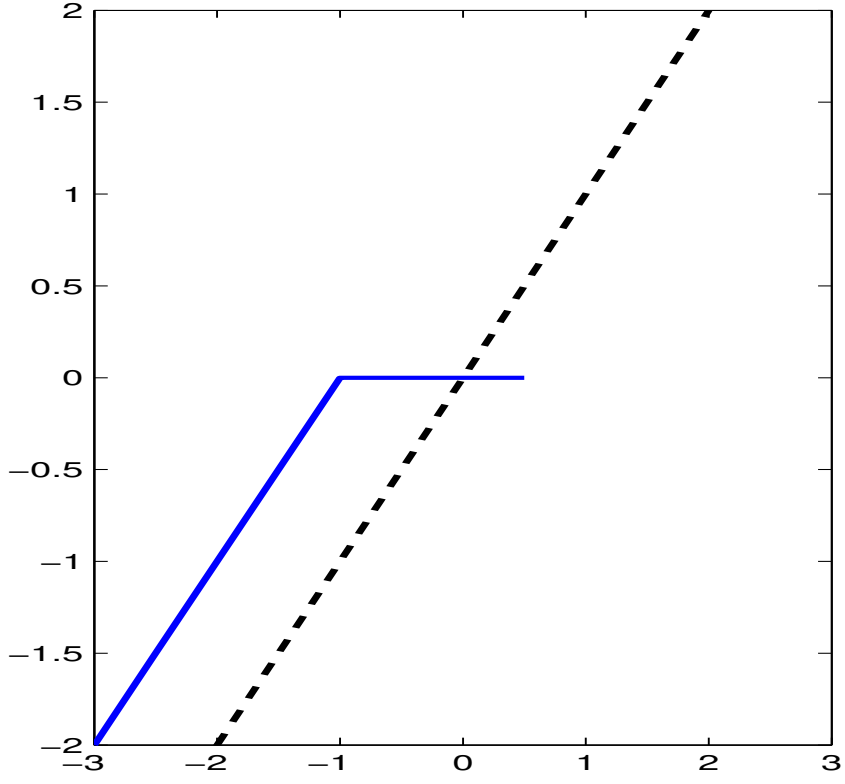
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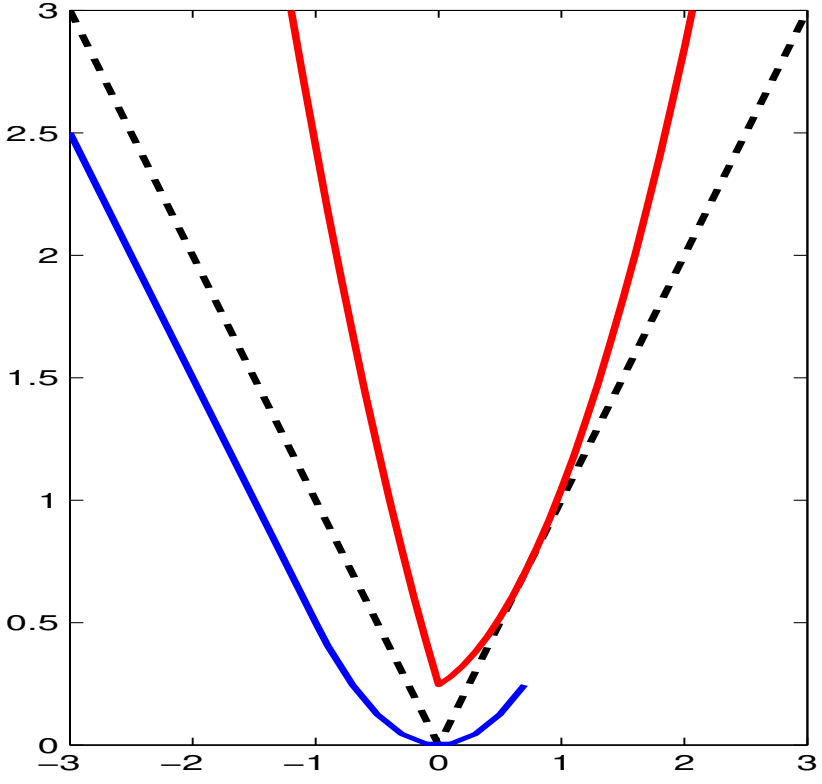
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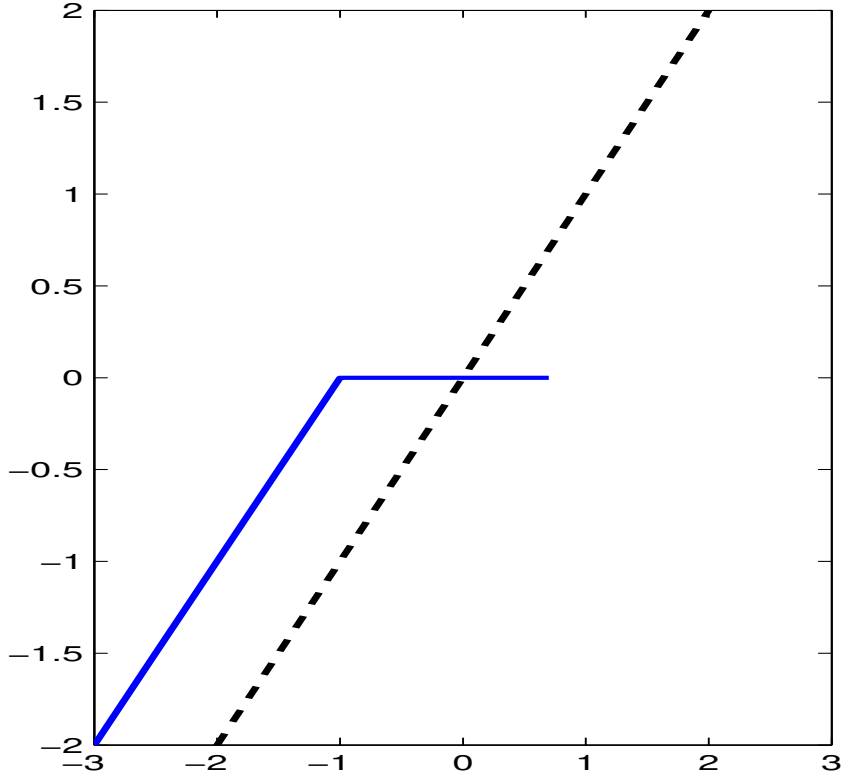
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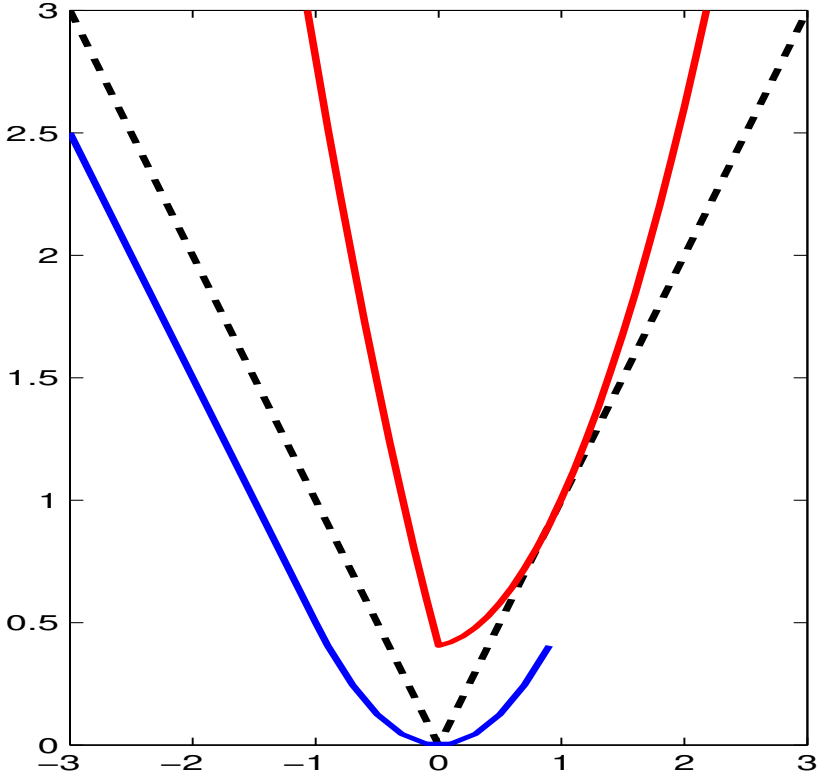
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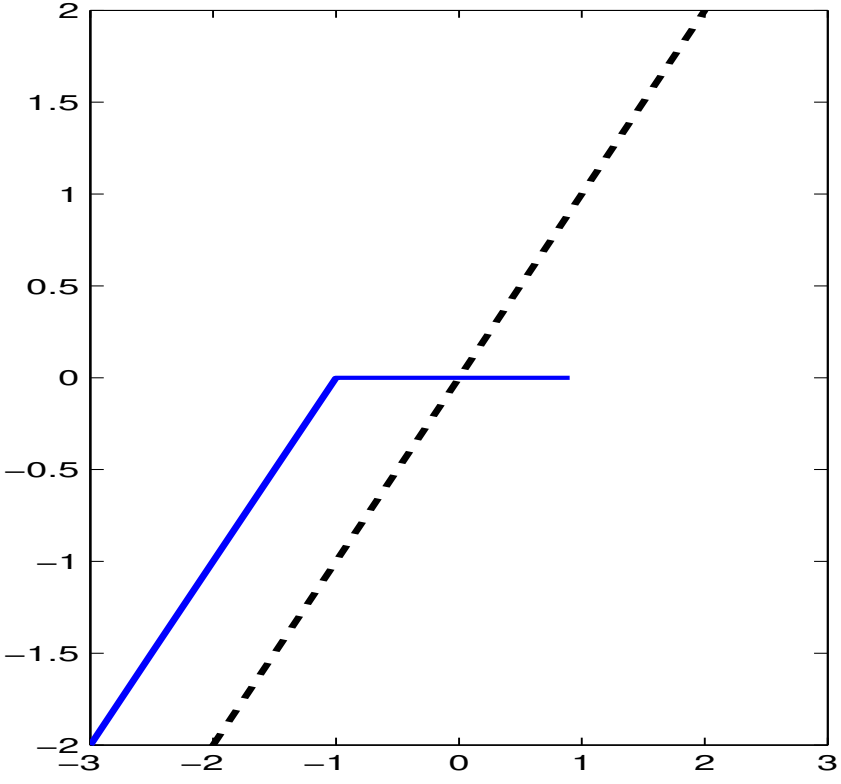
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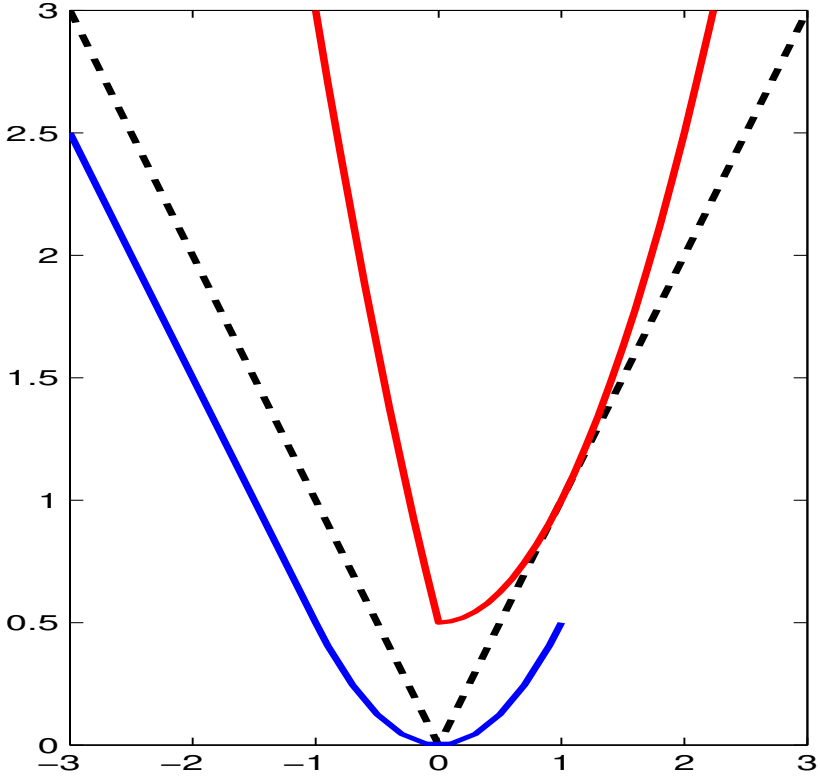
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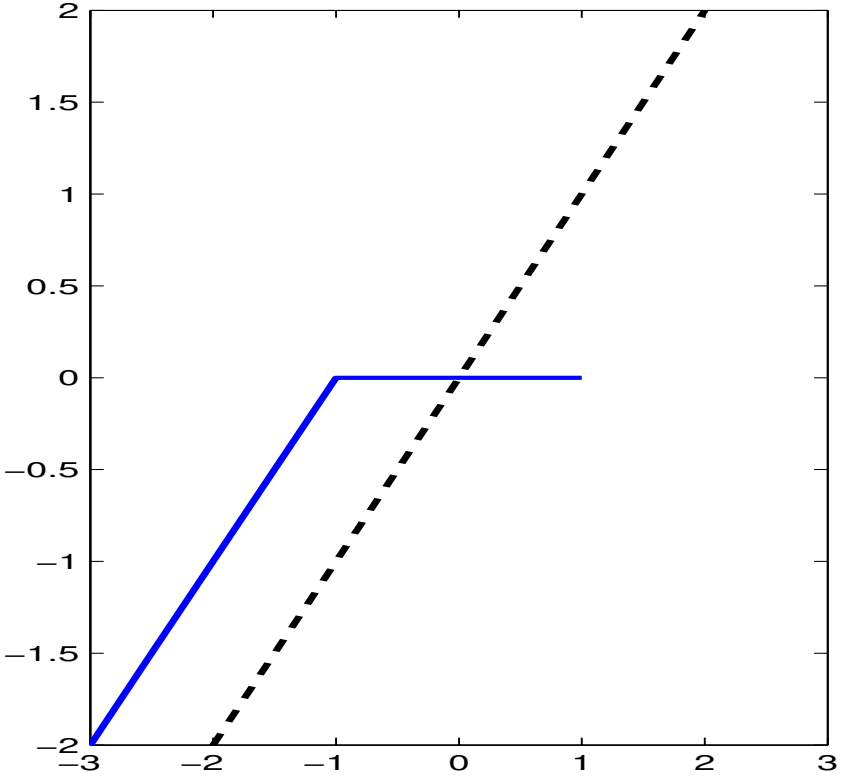
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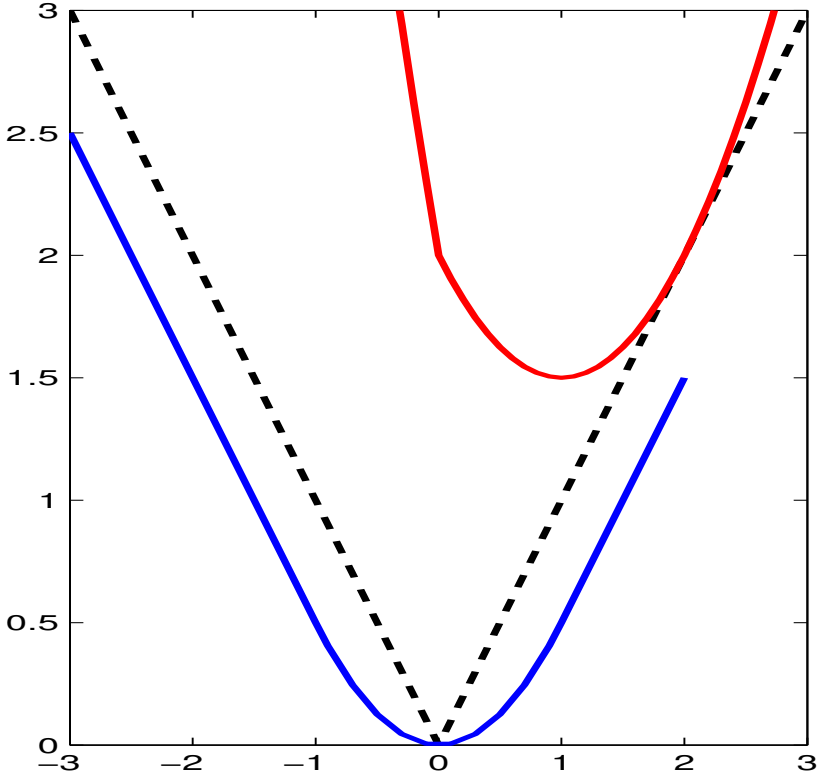
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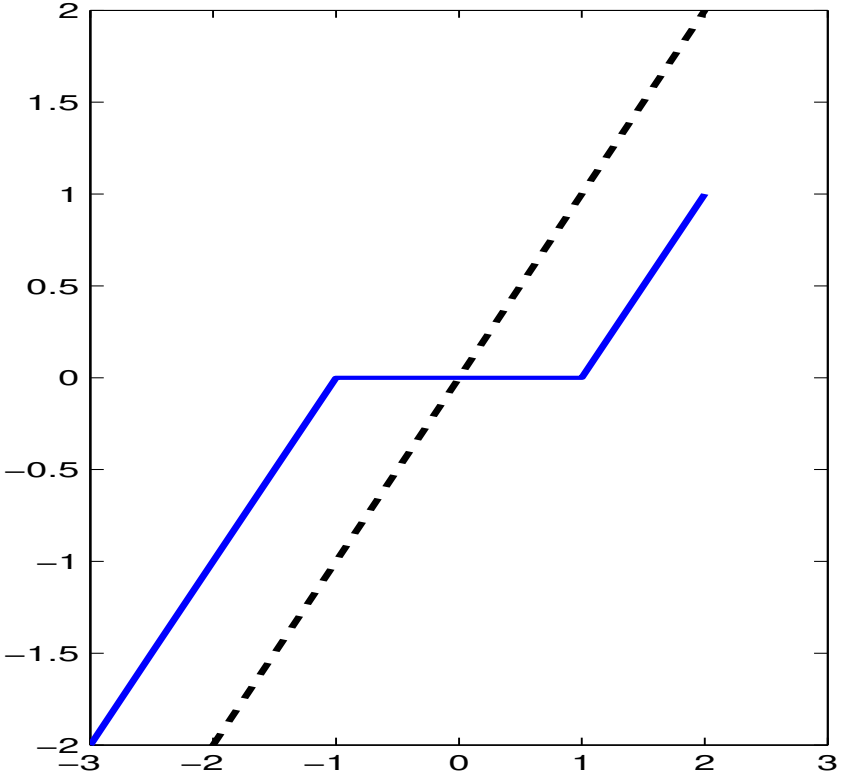
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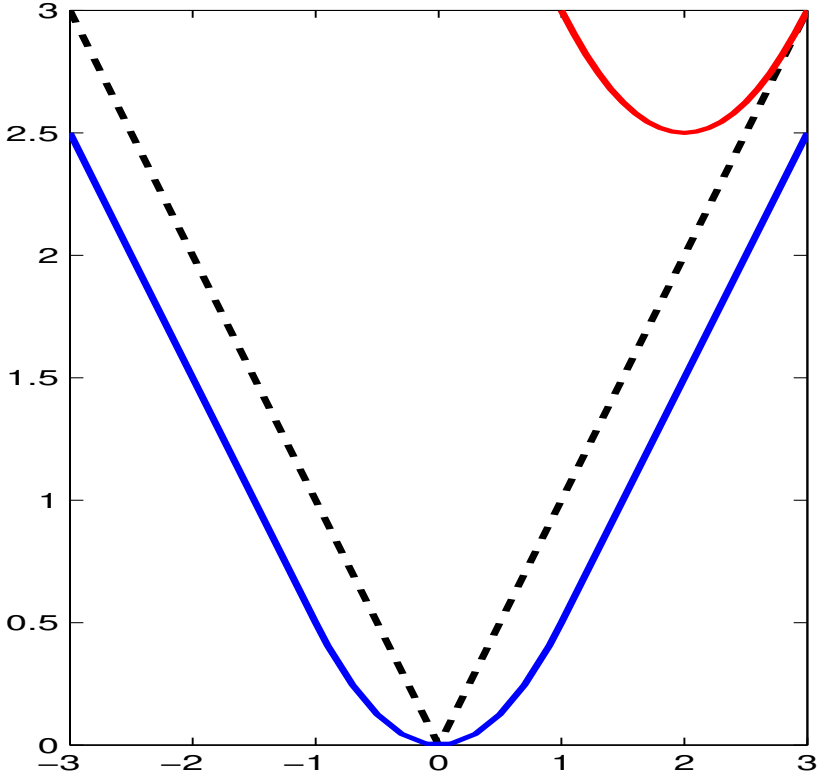
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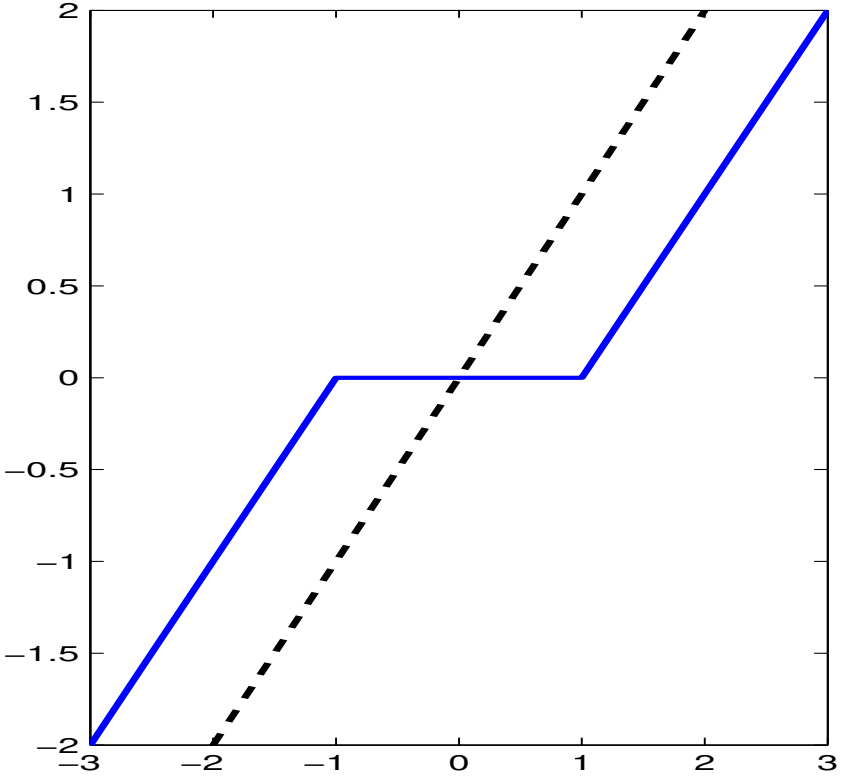
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Proximity operator: existence and uniqueness

Let $f \in \Gamma_0(\mathcal{H})$ and $\gamma \in \,]0, +\infty[$.
For every $x \in \mathcal{H}$, there exists a unique vector $p \in \mathcal{H}$ such that

$$f(p) + \frac{1}{2\gamma}\|p - x\|^2 = \inf_{y \in \mathcal{H}} f(y) + \frac{1}{2\gamma}\|y - x\|^2 \,.$$

Proof: $f \in \Gamma_0(\mathcal{H}) \Leftrightarrow f^* \in \Gamma_0(\mathcal{H})$. Thus, there exists $u \in \mathcal{H}$ such that $f^*(u) \in \mathbb{R}$. According to Fenchel-Young inequality, we have

$$(\forall y \in \mathcal{H}) \qquad f(y) \geq \langle u \mid y \rangle - f^*(u).$$

Then, $f(y) + (2\gamma)^{-1}\|y - x\|^2 \to +\infty$ when $\|y\| \to +\infty$.
Furthermore $(2\gamma)^{-1}\|\cdot - x\|^2$ being strictly convex, $f + (2\gamma)^{-1}\|\cdot - x\|^2$ is a strictly convex coercive function.

Proximity operator: characterization

Let $\mathcal{H}$ be a Hilbert space and $f \in \Gamma_0(\mathcal{H})$.

$$(\forall x \in \mathcal{H}) \quad p = \operatorname{prox}_f(x) \quad \Leftrightarrow \quad x - p \in \partial f(p) \,.$$

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Proof: By using Fermat's rule, for every $x \in \mathcal{H}$, $p = \operatorname{prox}_f(x)$ if and only if

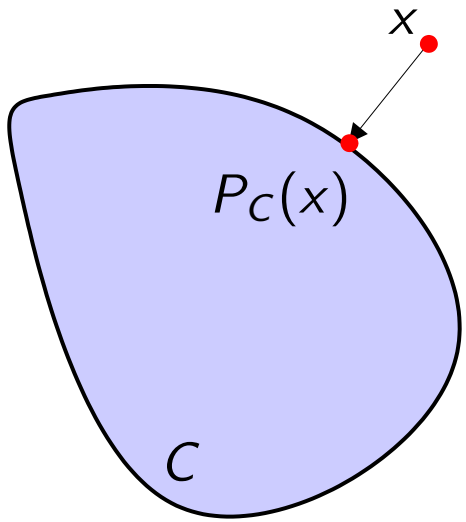
$$\begin{aligned} & p = \underset{y \in \mathcal{H}}{\arg\min} \; f(y) + \frac{1}{2}\|y - x\|^2 \\ \Leftrightarrow \quad & 0 \in \partial\Big(f + \frac{1}{2}\|\cdot - x\|^2\Big)(p) \\ \Leftrightarrow \quad & 0 \in \partial f(p) + p - x \\ \Leftrightarrow \quad & x \in (\mathrm{Id} + \partial f)(p). \end{aligned}$$

Proximity operator: examples

Projection :

Let $\mathcal{H}$ be a Hilbert space. Let C be a nonempty closed convex subset of $\mathcal{H}$.

$$(\forall x \in \mathcal{H}) \qquad \operatorname{prox}_{\iota_C}(x) = \operatorname*{argmin}_{y \in C} \frac{1}{2}\|y - x\|^2 = P_C(x).$$



Proximity operator: examples

Projection :
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Remark :

- $p = P_C(x) \Leftrightarrow x - p \in \partial \iota_C(p) = N_C(p)$
  $\Leftrightarrow$ $p \in C$ and $(\forall y \in C)\, \langle y - p \mid x - p \rangle \leq 0$.
  Particular case: if C is a vector space: $p = P_C(x) \Leftrightarrow \begin{cases} p \in C \\ x - p \in C^\perp \end{cases}$.
- ${}^{\gamma}\iota_C = (2\gamma)^{-1} d_C^2$ where d_C distance to the convex set C is defined by $d_C : x \mapsto \inf_{y \in C} \|y - x\| = \|x - P_C x\|$.

Proximity operator: examples

Power q function with $q \geq 1$:

Let $\chi > 0$, $q \in [1, +\infty[$ and $\varphi \colon \mathbb{R} \to]-\infty, +\infty]$: $\xi \mapsto \chi|\xi|^q$.
Then, for every $\xi \in \mathbb{R}$,

$$\operatorname{prox}_{\varphi}\xi = \begin{cases} \operatorname{sign}(\xi)\max\{|\xi| - \chi, 0\} & \text{if } q = 1 \\ \xi + \frac{4\chi}{3 \cdot 2^{1/3}}\left((\epsilon - \xi)^{1/3} - (\epsilon + \xi)^{1/3}\right) & \\ \qquad \text{where} \quad \epsilon = \sqrt{\xi^2 + 256\chi^3/729} & \text{if } q = \frac{4}{3} \\ \xi + \frac{9\chi^2 \operatorname{sign}(\xi)}{8}\left(1 - \sqrt{1 + \frac{16|\xi|}{9\chi^2}}\right) & \text{if } q = \frac{3}{2} \\ \frac{\xi}{1+2\chi} & \text{if } q = 2 \\ \operatorname{sign}(\xi)\frac{\sqrt{1+12\chi|\xi|}-1}{6\chi} & \text{if } q = 3 \\ \left(\frac{\epsilon+\xi}{8\chi}\right)^{1/3} - \left(\frac{\epsilon-\xi}{8\chi}\right)^{1/3} \quad \text{where} \quad \epsilon = \sqrt{\xi^2 + 1/(27\chi)} & \text{if } q = 4 \end{cases}$$

Proximity operator: examples

Plot the graphs of these proximity operator on the same figure.

Proximity operator: examples

Quadratic function :

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces.

Let $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$, $\gamma \in \,]0, +\infty[$ and $z \in \mathcal{G}$.

$$f = \gamma \left\| L \cdot - z \right\|^2 / 2 \quad \Rightarrow \quad \operatorname{prox}_f = (\mathrm{Id} + \gamma L^* L)^{-1} (\cdot + \gamma L^* z).$$

- Exercise : Prove this property.

Proximity operator: properties

Let $\mathcal{H}$ be a Hilbert space, $x \in \mathcal{H}$ and $f \in \Gamma_0(\mathcal{H})$.

Properties	$g(x)$	$\text{prox}_g x$
Translation	$f(x-z), z \in \mathcal{H}$	$z + \text{prox}_f(x-z)$
Quadratic perturbation	$f(x) + \alpha \parallel x \parallel^2 /2 + \langle z \mid x \rangle + \gamma$ $z \in \mathcal{H}, \alpha > 0, \gamma \in \mathbb{R}$	$\text{prox}_{\frac{f}{\alpha+1}}(\frac{x-z}{\alpha+1})$
Scaling	$f(\rho x), \rho \in \mathbb{R}^*$	$\frac{1}{\rho}\text{prox}_{\rho^2 f}(\rho x)$
Reflection	$f(-x)$	$-\text{prox}_f(-x)$
Moreau enveloppe	${}^{\gamma}f(x) = \inf_{y \in \mathcal{H}} f(y) + \frac{1}{2\gamma}\|x-y\|^2$ $\gamma > 0$	$\frac{1}{1+\gamma}\left(\gamma x + \text{prox}_{(1+\gamma)f}(x)\right)$

Proximity operator: properties

For every $i \in \{1, \ldots, n\}$, let $\mathcal{H}_i$ be a Hilbert space and let $f_i \in \Gamma_0(\mathcal{H}_i)$.
If

$$\big(\forall x = (x_1, \ldots, x_n) \in \mathcal{H}_1 \times \cdots \times \mathcal{H}_n\big) \quad f(x) = \sum_{i=1}^{n} f_i(x_i),$$

then

$$\big(\forall x = (x_1, \ldots, x_n) \in \mathcal{H}_1 \times \cdots \times \mathcal{H}_n\big) \quad \operatorname{prox}_f(x) = \big(\operatorname{prox}_{f_i}(x_i)\big)_{1 \leq i \leq n}.$$

Proximity operator: properties

Let $\mathcal{H}$ be a separable Hilbert space.
Let $(b_i)_{i\in I}$ be an orthonormal basis of $\mathcal{H}$.
For every $i \in I$, let $\varphi_i \in \Gamma_0(\mathbb{R})$ such that $\varphi_i \geq 0$. For every $x \in \mathcal{H}$, if

$$f(x) = \sum_{i\in I} \varphi_i(\langle x \mid b_i \rangle)$$

then

$$\operatorname{prox}_f(x) = \sum_{i\in I} \operatorname{prox}_{\varphi_i}(\langle x \mid b_i \rangle) b_i.$$

Remark: The assumption $(\forall i \in I)\ \varphi_i \geq 0$ can be relaxed if $\mathcal{H}$ is finite dimensional.

Proximity operator: properties

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then

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Example: $\mathcal{H} = \mathbb{R}^N$, $(b_i)_{1\leq i\leq N}$ canonical basis of $\mathbb{R}^N$, $f = \lambda\|\cdot\|_1$ with $\lambda \in [0, +\infty[$.

$$(\forall x = (x^{(i)})_{1\leq i\leq N} \in \mathbb{R}^N) \qquad \operatorname{prox}_{\lambda\|\cdot\|_1}(x) = \left(\operatorname{prox}_{\lambda|\cdot|}(x^{(i)})\right)_{1\leq i\leq N}$$

Proximity operator: properties

Moreau decomposition formula

Let $\mathcal{H}$ be a Hilbert space, $f \in \Gamma_0(\mathcal{H})$ and $\gamma \in]0, +\infty[$.

$$(\forall x \in \mathcal{H}) \qquad \operatorname{prox}_{\gamma f^*} x = x - \gamma \operatorname{prox}_{\gamma^{-1} f}(\gamma^{-1} x) \,.$$

Proof:

$$\begin{aligned}
p = \operatorname{prox}_{\gamma f^*} x &\Leftrightarrow x - p \in \gamma \partial f^*(p) \\
&\Leftrightarrow p \in \partial f\left(\frac{x-p}{\gamma}\right) \\
&\Leftrightarrow \frac{x}{\gamma} - \frac{x-p}{\gamma} \in \frac{1}{\gamma} \partial f\left(\frac{x-p}{\gamma}\right) \\
&\Leftrightarrow \frac{x-p}{\gamma} = \operatorname{prox}_{\gamma^{-1} f}(\gamma^{-1} x) \\
&\Leftrightarrow p = x - \gamma \operatorname{prox}_{\gamma^{-1} f}(\gamma^{-1} x).
\end{aligned}$$

Proximity operator: properties

Moreau decomposition formula

Let $\mathcal{H}$ be a Hilbert space, $f \in \Gamma_0(\mathcal{H})$ and $\gamma \in \,]0, +\infty[$.

$$(\forall x \in \mathcal{H}) \qquad \operatorname{prox}_{\gamma f^*} x = x - \gamma \operatorname{prox}_{\gamma^{-1} f}(\gamma^{-1} x)\,.$$

Example: If $\mathcal{H} = \mathbb{R}^N$, $f = \frac{1}{q}\|\cdot\|_q^q$ with $q \in]1, +\infty[$, then $f^* = \frac{1}{q^*}\|\cdot\|_{q^*}^{q^*}$ with $1/q + 1/q^* = 1$, and

$$(\forall x \in \mathbb{R}^N) \qquad \operatorname{prox}_{\frac{\gamma}{q^*}\|\cdot\|_{q^*}^{q^*}} x = x - \gamma \operatorname{prox}_{\frac{1}{\gamma q}\|\cdot\|_q^q}(\gamma^{-1} x).$$

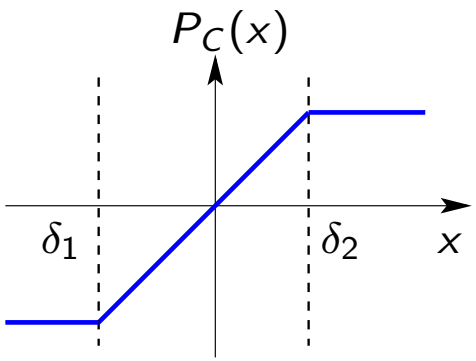
Proximity operator: examples

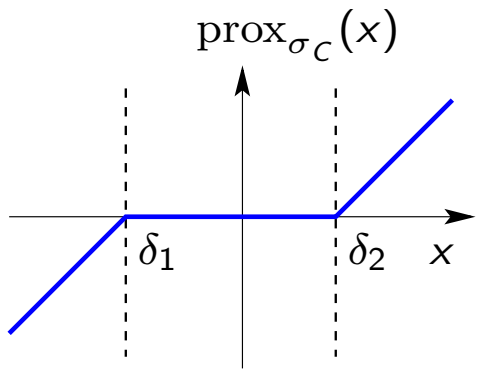
Support function :
Let $\mathcal{H}$ be a Hilbert space and $C \subset \mathcal{H}$ be nonempty closed convex.

$$(\forall x \in \mathcal{H}) \qquad \operatorname{prox}_{\sigma_C} = \mathrm{Id} - P_C.$$

Soft-thresholding : $\mathcal{H} = \mathbb{R}$, $\delta_1 = \inf C$ and $\delta_2 = \sup C$. For every $x \in \mathbb{R}$,

$$\sigma_C(x) = \begin{cases} \delta_1 x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ \delta_2 x & \text{if } x > 0 \end{cases} \quad \Rightarrow \quad \operatorname{prox}_{\sigma_C}(x) = \operatorname{soft}_C(x) = \begin{cases} x - \delta_1 & \text{if } x < \delta_1 \\ 0 & \text{if } x \in C \\ x - \delta_2 & \text{if } x > \delta_2. \end{cases}$$





Proximity operator: properties

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces. Let $f \in \Gamma_0(\mathcal{H})$ and $L \in \mathcal{B}(\mathcal{G}, \mathcal{H})$ such that $LL^* = \mu \mathrm{Id}$ where $\mu \in \,]0, +\infty[$. Then

$$\mathrm{prox}_{f \circ L} = \mathrm{Id} - \mu^{-1} L^* \circ (\mathrm{Id} - \mathrm{prox}_{\mu f}) \circ L.$$

Proximity operator: properties

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$$\mathrm{prox}_{f \circ L} = \mathrm{Id} - \mu^{-1} L^* \circ (\mathrm{Id} - \mathrm{prox}_{\mu f}) \circ L.$$

Proof: $LL^* = \mu \mathrm{Id} \Rightarrow \mathrm{ran}\, L = \mathcal{H}$ is closed, hence
$V = \mathrm{ran}\,(L^*) = (\ker L)^\perp$ is closed. The orthogonal projection onto V is $P_V = L^*(LL^*)^{-1}L = \mu^{-1}L^*L$.
For every $x \in \mathcal{H}$, $p = \mathrm{prox}_{f \circ L} x \Leftrightarrow x - p \in \partial(f \circ L)(p) = L^* \partial f(Lp)$ (since $\mathrm{ran}\, L = \mathcal{H}$). Thus, $x - p \in V$.
It can be deduced that $P_{V^\perp} p = P_{V^\perp} x = x - P_V x = x - \mu^{-1} L^* L x$.
Furthermore,
$x - p \in L^* \partial(Lp) \Rightarrow Lx - Lp \in \mu \partial f(Lp) \Leftrightarrow Lp = \mathrm{prox}_{\mu f}(Lx)$.
We have thus $P_V p = \mu^{-1} L^* Lp = \mu^{-1} L^* \mathrm{prox}_{\mu f}(Lx)$ and
$p = P_V p + P_{V^\perp} p = x - \mu^{-1} L^* (\mathrm{Id} - \mathrm{prox}_{\mu f})(Lx)$.